

When Lines Become Contagious: Analyzing the Spread of Infection in the M/M/1 Queue

Laurel Newman, Jamol Pender

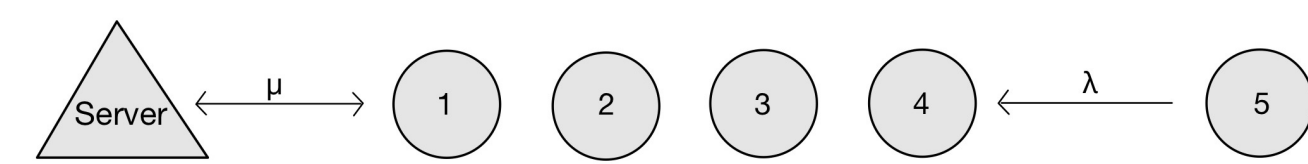
Stochastic Disease Models

Compartment models (e.g., SEIR model): individuals move between Susceptible, Exposed, Infectious, and Recovered states.

Population heterogeneity in disease modelling: trade-off between practical accuracy from simulations vs. theoretical rigor.

Mathematical Background

M/M/1 queues: system with a single server, sequential customers, arrival times generated from a Poisson process with parameter λ , service times exponential with parameter μ .



Customer overlaps: in service systems, the entire population does not interact simultaneously.

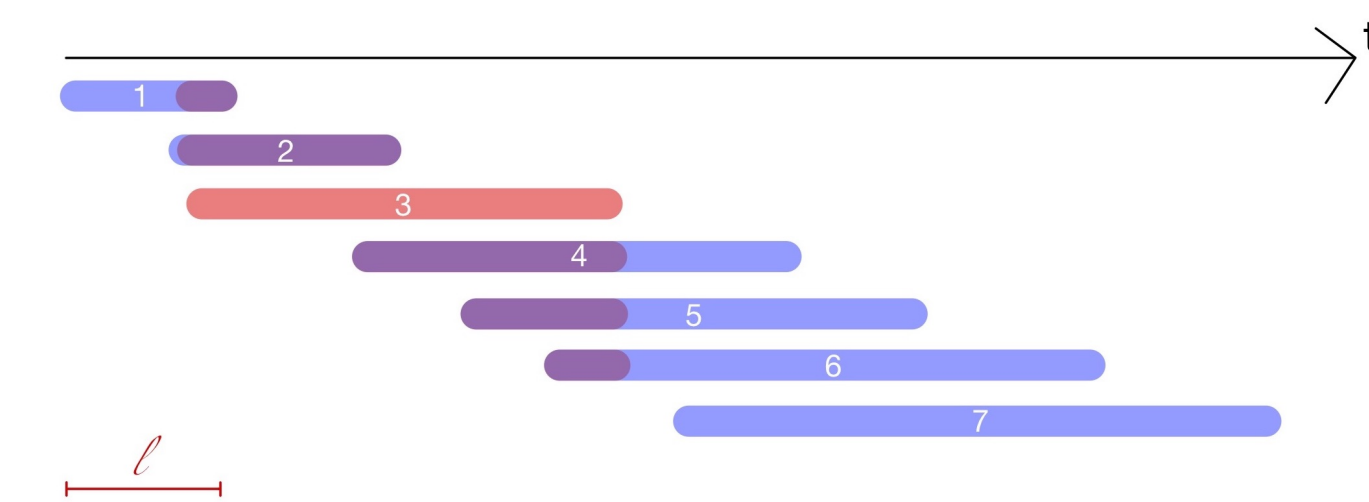


Figure 1. An overlap of at least ℓ with an infectious customer infects you!

Key Insight

Customers arrive infectious with probability p , and susceptible otherwise.

Dual perspectives: individual (people who infect us) and communal (people we infect):



Figure 2. Susceptible customer perspective.

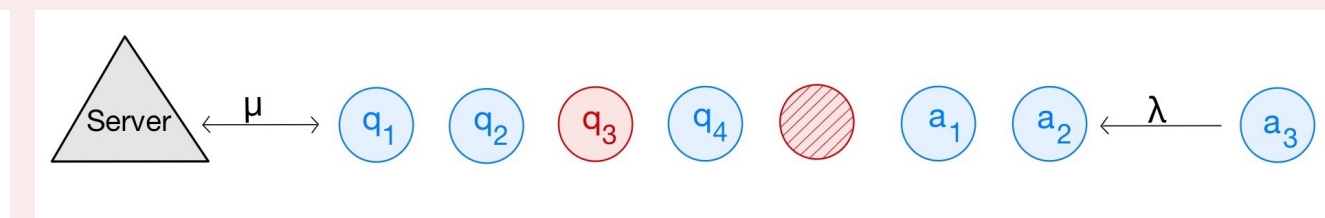


Figure 3. Infectious customer perspective.

Partitioning Approach

Partition into customers present in the queue upon our arrival, and the arrivals after us.

Our dual perspectives, see Figures 2 and 3, allow us to immediately simplify after arrivals.

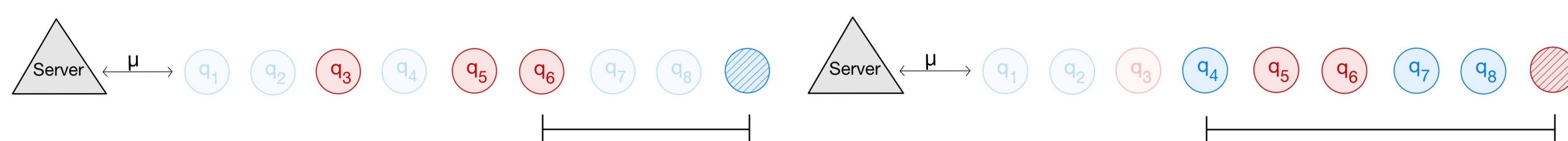


Figure 4. For a susceptible customer, considering upon arrivals reduces to considering closest infectious customer.

Figure 5. For an infectious customer, considering upon arrivals reduces to considering first susceptible customer who overlaps at least ℓ .

Theorem 1: Susceptible customer infection probability

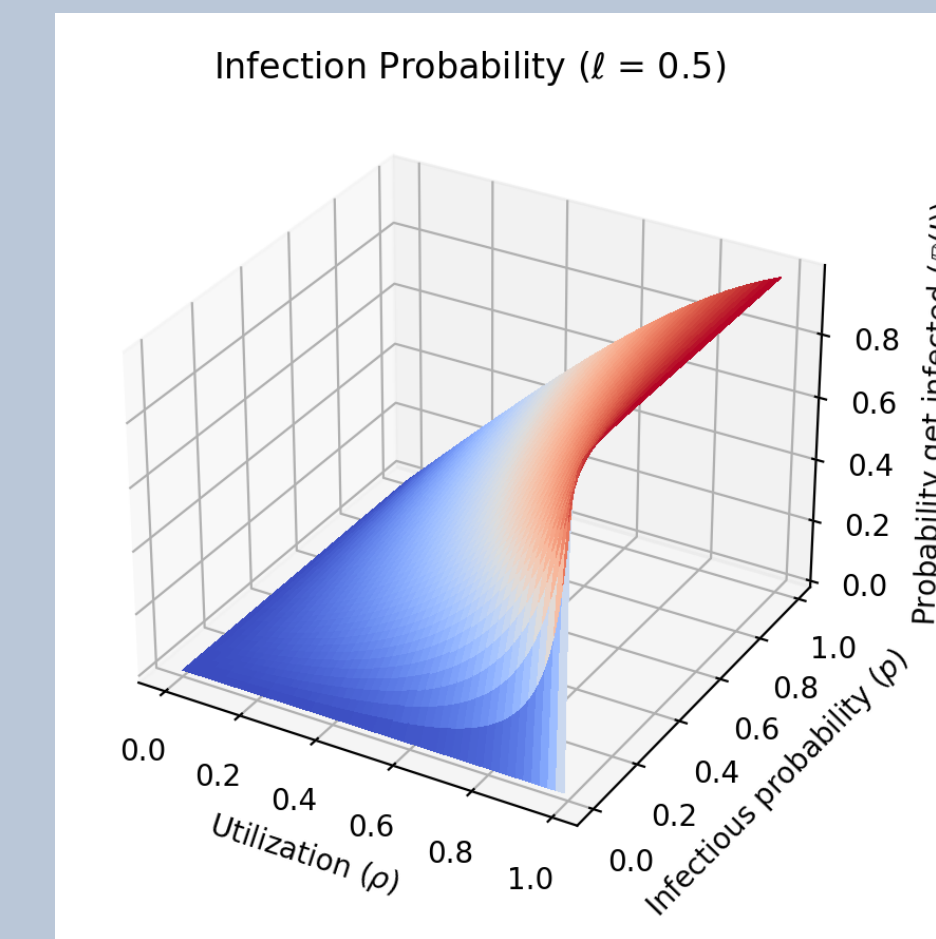
Let I be the event a customer gets infected while in the queue. Suppose a susceptible customer joins the $M/M/1$ queue in steady state, then the probability that the customer gets infected is equal to,

$$\mathbb{P}(I) = \frac{2\lambda p}{\mu - \lambda + 2\lambda p} e^{-(\mu - \lambda)\ell}.$$

The probability of infection is the product of two probabilities:

- sojourn time is at least infection threshold,
- and the exponential of our pass-through rate is more than the exponential of twice the infectious arrival rate.

Twice the rate accounts for the equal probability of infection in both the upon and after arrival cases.



Theorem 2: Moments of number infectious customer infects

Suppose an infectious customer joins the $M/M/1$ queue in steady state, then the m^{th} moment of the number of susceptible customers they infect is,

$$\mathbb{E}[N_I^m] = \frac{\mu - \lambda}{\mu - \lambda + 2\lambda(1 - p)} \text{Li}_{-m} \left(\frac{2\lambda(1 - p)}{\mu - \lambda + 2\lambda(1 - p)} \right),$$

where $\text{Li}_m(x)$ is the polylogarithm function and is defined by

$$\text{Li}_m(x) = \sum_{j=1}^{\infty} \frac{x^j}{j^m}.$$

Thus, the mean and variance of the number infected by one infectious customer is equal to

$$\mathbb{E}[N_I] = \frac{2\lambda(1 - p)}{\mu - \lambda} e^{-(\mu - \lambda)\ell}, \text{ and}$$

$$\text{Var}[N_I] = \frac{2\lambda(1 - p)(\mu - \lambda + 4\lambda(1 - p))}{(\mu - \lambda)^2} e^{-(\mu - \lambda)\ell} - \frac{4\lambda^2(1 - p)^2}{(\mu - \lambda)^2} e^{-2(\mu - \lambda)\ell}.$$

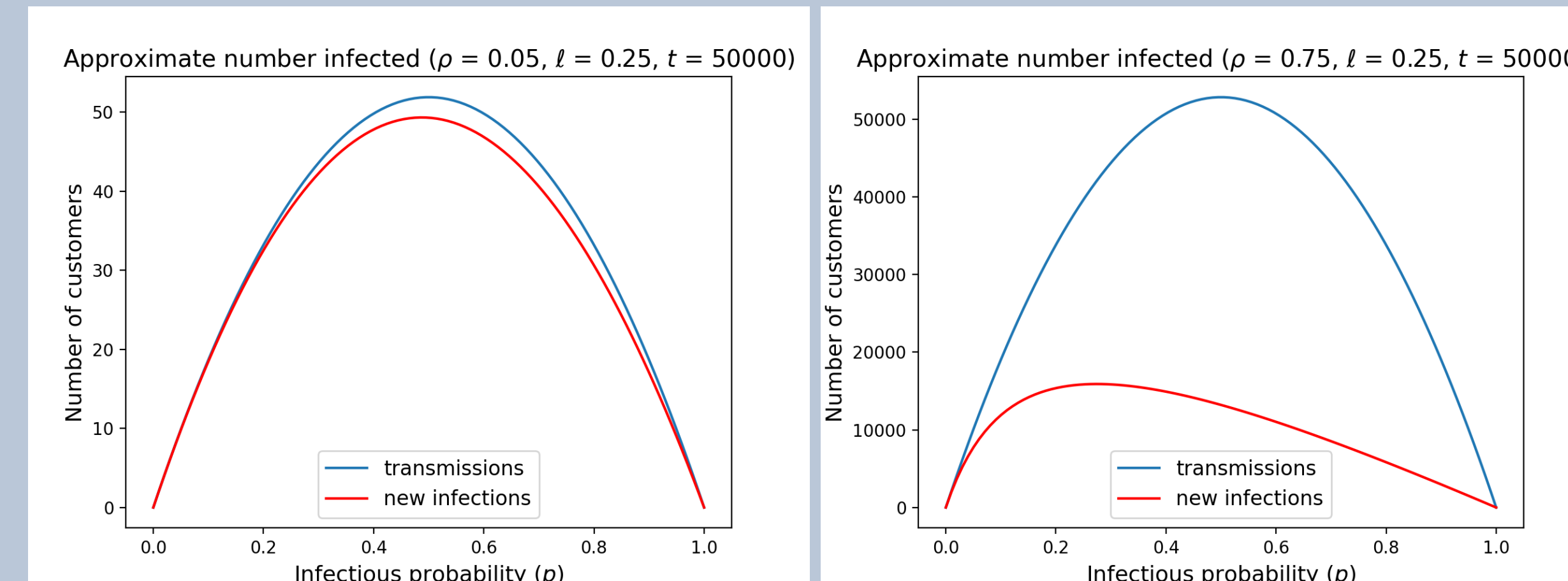


Figure 6. Higher server utilization, $\rho = \frac{\lambda}{\mu}$, causes these formulas to diverge.

Numerical Results

λ	p	$\mathbb{P}(I)$ (formula)	$\mathbb{P}(I)$ (simulation)
.50	.01	.01527	.01578
.75	.01	.04995	.04941
.90	.01	.14510	.14885
.50	.02	.02995	.03023
.75	.02	.09455	.09473
.90	.02	.25180	.24421

λ	p	$\mathbb{P}(I)$ (formula)	$\mathbb{P}(I)$ (simulation)
.50	.10	.12980	.12893
.75	.10	.33094	.33323
.90	.10	.61150	.60601
.50	.50	.38940	.38959
.75	.50	.66187	.66032
.90	.50	.85611	.85904

Table 1. For a susceptible arrival in steady state, the theoretical and simulated probabilities that this customer is infected. We consider 100,000 customers per each simulation.

λ	p	$\mathbb{E}[N_I]$ (form.)	$\mathbb{E}[N_I]$ (sim.)	$\text{Var}[N_I]$ (form.)	$\text{Var}[N_I]$ (sim.)
.50	.01	1.542	1.573	5.271	5.422
.75	.01	5.242	5.274	40.038	39.889
.90	.01	16.951	17.354	333.748	371.666
.50	.02	1.526	1.532	5.180	5.259
.75	.02	5.189	5.225	39.286	40.449
.90	.02	16.780	16.780	327.209	317.466

Table 2. For an infectious arrival in steady state, the theoretical and simulated mean and variance of the number of customers they infect. We consider 500,000 customers per each simulation.

Extensions

M/G/∞ Queues:

Infinite number of servers

General service times distribution

- Susceptible customer infection probability
- Higher moments of number infected

M/G/∞ Comorbidity:

Multiple diseases in the system

- Probability of contracting a subset
- Probability of contracting only that subset

Cybersecurity:

Side-channel attacks

- After arrival probability
- Overlapping sufficiently with service

In-System Infectiousness:

Become infectious after getting infected

Different Service Regimes:

Depending on prevalence of infection in system, prioritize serving infectious or susceptible people first.

References

- Kang Kang, Sherwin Doroudi, Mohammad Delasay, and Alexander Wickehham. A queueing-theoretic framework for evaluating transmission risks in service facilities during a pandemic. *Production and Operations Management*, 2022.
- Edward H Kaplan. OM forum—COVID-19 scratch models to support local decisions. *Manufacturing & Service Operations Management*, 22(4):645–655, 2020.
- Sergio D Palomo and Jamol Pender. Overlap Times in the $GI^B/GI/\infty$ Queue. *arXiv preprint arXiv:2302.07410*, 2023.